

NIFTY-50 Investment Intelligence Platform

Technical Report

Dataset: NIFTY-50 Stock Market Data (Kaggle), 2000–2021

Stack: Python · pandas · scikit-learn · XGBoost · SciPy · SHAP · Streamlit

Repository: <https://gitlab.com/rakshit11147-group/nifty-last>

1. Exploratory Data Analysis (EDA)

The dataset contains daily OHLCV (Open, High, Low, Close, Volume) records for all NIFTY-50 constituents from 2000 to 2021, combined in *NIFTY50_all.csv*.

Steps performed (notebooks/01_EDA.PY):

- **Data quality checks:** missing values, duplicate dates, and inconsistent symbols were identified and cleaned. Dates were parsed and set as a sorted index per stock.
- **Price trends:** long-term close-price trajectories were plotted per stock, showing strong secular growth in IT (TCS, INFY) and banking (HDFCBANK) names, and high cyclicalities in metals and energy.
- **Return distributions:** daily returns are non-normal, exhibiting fat tails and negative skew. Extreme moves cluster around crisis periods (2008 global financial crisis, 2020 COVID crash).
- **Volatility clustering:** rolling 20-day volatility confirms that high-volatility periods cluster in time, motivating volatility-based features.
- **Correlation analysis:** a cross-stock return correlation heatmap shows strong intra-sector correlation (banks with banks, IT with IT), which justifies diversification across sectors in portfolio construction.

Key takeaway: prices are non-stationary, but returns and indicator-based ratios are usable as model inputs. Direction (up/down) is a more learnable target than the exact price level.

2. Feature Engineering

All features are computed in *src/utis/indicators.py* (`TechnicalIndicators.add_all()`), configured through *config/settings.py*.

Feature group	Features	Purpose
Trend	SMA 20/50/200, EMA 12/26, Price_vs_SMA20/50	Trend position relative to moving averages
Momentum	MACD, MACD_Signal, MACD_Hist, RSI_14, ROC_10, Momentum_5/20	Speed and persistence of moves

Feature group	Features	Purpose
Volatility	Bollinger %B and Width, ATR_14 (% of price), rolling 20-day vol	Regime and range information
Volume	OBV (On-Balance Volume)	Volume confirmation of price moves
Price structure	Daily log return, High/Low ratio, Open/Close ratio	Intraday behavior

Design choices:

- Ratios and normalized features (e.g., $\text{Price_vs_SMA20} = \text{Close/SMA20} - 1$, ATR_Pct) are used instead of raw prices so features are comparable across stocks and time, and stationary enough for tree models.
- Rows with NaN values (from rolling windows) are dropped after computation.
- The final model feature set contains **15 features** (StockPredictor.FEATURE_COLS).

3. Methodology

The overall pipeline is:

```
Raw OHLCV → Cleaning → Technical indicators → Feature matrix X → Label: direction
(1 if Close[t+h] > Close[t], else 0) → Chronological train/test split (80/20, no
shuffle) → Train RF + XGBoost → Evaluate → Explain (SHAP / importances)
```

- **Prediction target:** binary direction of the close price h days ahead (default h = 1). This frames forecasting as classification, which is more robust than exact price regression on noisy financial data.
- **Train/test split:** strictly chronological (first 80% train, last 20% test) to avoid look-ahead bias. Data is never shuffled.
- **Feature scaling:** MinMax scaling applied where required.
- **Evaluation metrics:** accuracy and per-class precision/recall/F1 (classification report) for direction models; MAE, RMSE, and R² for price-level forecasts.
- **Persistence:** trained models are serialized with joblib to models_saved/.

4. Model Architecture

Implemented in *src/models/predictor.py* (StockPredictor class).

Model	Task	Key hyperparameters
Random Forest	Direction classification	200 trees, max_depth = 10, random_state = 42

Model	Task	Key hyperparameters
XGBoost	Direction classification	200 estimators, max_depth = 6, learning_rate = 0.05, subsample = 0.8
Ensemble	Direction	Agreement / averaged probability of RF and XGBoost

- Both models consume the same 15-feature matrix and binary direction label.
- XGBoost's regularized gradient boosting captures nonlinear feature interactions; Random Forest provides a stable, lower-variance baseline. Combining them reduces single-model risk.
- The architecture also supports a price-level regressor for multi-day price paths used by the dashboard's forecast view.
- The engine is symbol-agnostic: one model instance is trained per stock (e.g., `StockPredictor(symbol='RELIANCE')`).

5. Portfolio Construction Logic

Implemented in `src/models/portfolio.py` (PortfolioBuilder class) using **mean-variance optimization** solved with SciPy's SLSQP optimizer.

Inputs: daily close prices of the selected stocks → daily returns → annualized mean return vector and covariance matrix (252 trading days). Risk-free rate: **6%** (India 10Y G-Sec proxy).

Three investor profiles, three objectives:

Profile	Objective	Target return	Max volatility
Conservative	Minimize portfolio volatility	8%	12%
Balanced	Maximize Sharpe ratio	15%	20%
Aggressive	Maximize expected return	25%	35%

Constraints applied to every optimization:

- Weights sum to 1 (fully invested)
- Per-stock bounds of **2% to 40%** to enforce diversification and prevent corner solutions
- Long-only (no short selling)

6. Risk Assessment Methodology

Implemented in `src/models/risk.py` (RiskAssessor class). All metrics are annualized with 252 trading days and a 6% risk-free rate.

Metric	Definition
Historical Volatility	Std. dev. of daily returns x sqrt(252)
Sharpe Ratio	(Annual return - risk-free) / volatility
Sortino Ratio	(Annual return - risk-free) / downside volatility only
Maximum Drawdown	Largest peak-to-trough decline of the cumulative value
VaR (95%)	Worst expected daily loss at 95% confidence; historical (empirical 5th percentile) and parametric (normal z-score) methods
CVaR / Expected Shortfall	Mean loss in the tail beyond VaR
Beta	Sensitivity vs. the NIFTY-50 index benchmark
Calmar Ratio	Annual return / max drawdown

7. Explainability Techniques

- **SHAP (SHapley Additive exPlanations):** TreeExplainer is applied to the trained tree models to quantify each feature's contribution to individual predictions and global importance, answering *why* the model predicts up or down for a given day.
- **Feature importances:** native Random Forest and XGBoost gain-based importances are compared with SHAP rankings for consistency.
- **Transparent metrics:** every risk and portfolio figure shown in the UI maps to a documented formula in risk.py/portfolio.py, so results are auditable rather than black-box.

Across stocks, momentum (RSI, MACD histogram) and short-term volatility features consistently rank as the strongest drivers of next-day direction predictions.

8. Key Insights

1. Direction beats price.

Predicting up/down direction yields meaningfully better-than-chance accuracy, while exact price-level forecasting remains noisy; the platform therefore presents probabilities, not certainties.

2. Momentum and volatility dominate.

RSI, MACD histogram, and rolling volatility are the most influential features per SHAP, confirming short-horizon mean-reversion/momentum effects in NIFTY-50 stocks.

3. Diversification is measurable.

Optimized portfolios materially reduce volatility versus single stocks at comparable returns; the Conservative profile cuts volatility well below the index while preserving most of the return.

4. Risk profiles matter.

The same stock universe produces very different allocations across Conservative/Balanced/Aggressive profiles, demonstrating that investor constraints, not just expected returns, drive allocation.

5. Fat tails are real.

Historical VaR consistently exceeds parametric VaR in magnitude, confirming that normal-distribution assumptions understate Indian equity tail risk; CVaR is reported for a more honest tail estimate.

6. Honest evaluation.

Chronological splits, no shuffling, and out-of-sample testing were enforced throughout to avoid look-ahead bias, a common pitfall in financial ML.

End of report.